E-Commerce Return Rate Investigation

AI & ML Internship – Task 05

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Internship: AI & ML Internship

Task Number: Task 05

Project Title: E-Commerce Return Rate Investigation

1. Abstract

This project analyzes an e-commerce dataset to understand product return behavior. Statistical analysis and data visualization techniques were used to identify return patterns, detect outliers, perform hypothesis testing, and generate business recommendations. The analysis helps organizations understand the reasons behind product returns and improve customer satisfaction.

2. Problem Statement

Product returns are one of the major challenges in the e-commerce industry. High return rates increase logistics costs, warehouse workload, inventory issues, and financial losses. The objective of this project is to analyze return behavior and identify the factors contributing to product returns.

3. Objective

- To analyze customer return behavior.
- To perform statistical analysis on the dataset.
- To identify high-return product categories.
- To detect outliers using statistical methods.
- To perform hypothesis testing.
- To provide business recommendations.

4. Dataset Description

Dataset Name: Ecommerce_Return_Rate_Dataset_500Rows.csv

Number of Records: 500

Number of Columns: 25

Returned Products: 17

Return Percentage: 3.4%

The dataset contains order information such as product category, quantity, amount, order status, shipping details, and return rate.

5. Tools and Technologies Used

- Python
- Google Colab
- GitHub

Python Libraries Used:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- SciPy

6. Methodology

Phase 1 – Data Understanding

The dataset was loaded using Pandas. Missing values, duplicate records, column names, data types, and dataset dimensions were analyzed.

Completed Tasks:

- Loaded Dataset
- Checked Shape
- Checked Columns
- Checked Missing Values

- Removed Duplicates
- Calculated Total Orders
- Calculated Return Percentage

Phase 2 – Descriptive Statistical Analysis

The following statistical measures were calculated:

- Mean
- Median
- Standard Deviation
- Quartiles

These statistics helped understand the distribution of the numerical data.

Phase 3 – Return Pattern Investigation

Category-wise return rate analysis was performed.

Visualizations created:

- Bar Chart
- Histogram
- Boxplot
- Heatmap

Observation: The Western Dress category recorded the highest return rate among all product categories.

Phase 4 – Outlier Analysis

Outliers were identified using:

- Boxplot
- IQR Method

High-priced products were identified as outliers, indicating records that require further business investigation.

Phase 5 – Hypothesis Testing

A Chi-Square Test was performed to determine whether Product Category and Order Status are statistically related.

The p-value was used to interpret whether the null hypothesis should be accepted or rejected.

Phase 6 – Customer Segmentation

Customers were segmented into:

- Normal Customers
- Premium Customers
- High Return Customers

A bar chart was created to visualize customer segmentation.

Phase 7 – Business Insights

The analysis identified the following insights:

- Western Dress products have the highest return rate.
- High-value products should be monitored carefully.
- Product descriptions and size charts should be improved.
- Better quality inspection can reduce returns.
- Customer feedback should be collected and analyzed regularly.

7. Results

- Total Orders: **500**
- Returned Products: **17**
- Return Percentage: **3.4%**
- Highest Return Category: **Western Dress**
- Statistical Analysis Completed Successfully
- Visualizations Generated Successfully

8. Conclusion

This project successfully analyzed product return behavior using statistical analysis and visualization techniques. The findings help businesses understand return patterns and support better operational decisions. The project also improved practical knowledge of data analysis using Python.

9. Future Scope

- Include Delivery Time analysis.
- Include Customer Rating analysis.
- Include Discount analysis.
- Build a Machine Learning model to predict product returns.
- Develop an interactive dashboard using Power BI or Tableau.

10. References

1. Amazon Sale Report Dataset
2. Python Documentation
3. Pandas Documentation
4. NumPy Documentation
5. Matplotlib Documentation
6. Seaborn Documentation
7. SciPy Documentation
8. Google Colab
9. GitHub